

CSCI1470: Deep Learning

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Storytelling with DL: An Exploration of Image-to-Story Generation

Github Repo: <https://github.com/Robert-Daly-256/CSCI1470-Final>

Presentation Slides:

<https://docs.google.com/presentation/d/1Unc2mC99L9bFuT3emoFCIShJs-gHv1mkoqtZZBgWn4E/edit?usp=sharing>

Introduction

Our project explores the automatic generation of short stories from input images. While many tools exist to convert stories into images, there are fewer that reverse this process, making our project a novel exploration in the field of visual-to-text storytelling. Our motivation stems from discussions about enhancing storytelling and language learning through digital picture books. Individuals are more likely to recall stories when they are accompanied by images, a psychological phenomenon known as the Pictorial Superiority Effect (Nelson et. al. 1976). This effect suggests that visual and verbal information encode more powerfully in memory than one alone, making images powerful tools for enhancing learning and recognition. This underscores the potential impact of visual storytelling systems, especially in educational applications. Storytelling and image captioning are both active areas of deep learning investigation, and combining them presents new possibilities for generating narratives from visual inputs.

Methodology

Preprocessing/Data

We train two models. As such, we preprocess two datasets.

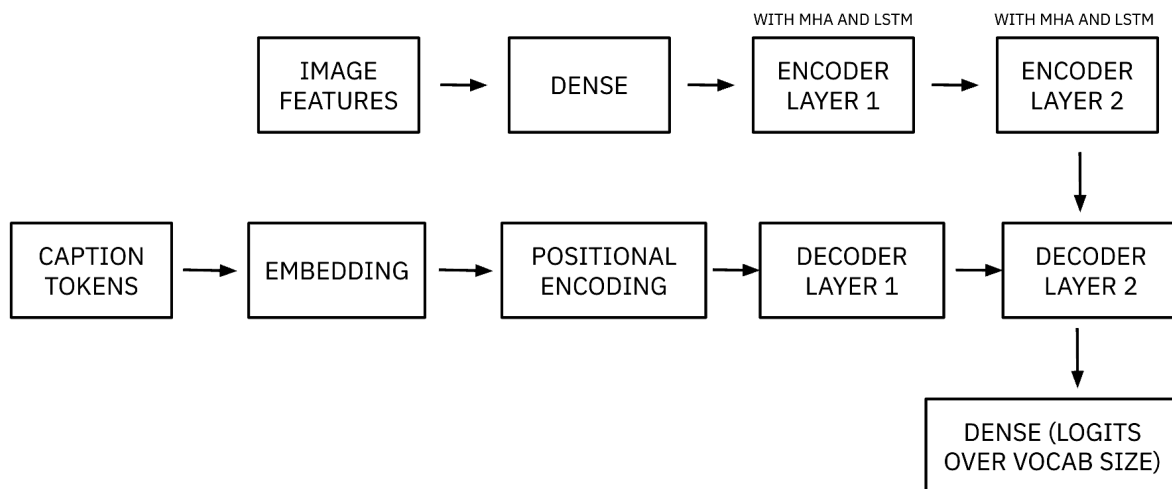
For our first dataset, we're preprocessing the Flickr8k dataset to create an image to prompt dataset. We resize the images and normalize them to the range $[0, 1]$ before feeding them into a ResNet model with pre-trained imagenet weights.

The prompt to story dataset is more complex. The dataset is webscraped from Reddit. We tokenize the text based on standard separation (\t, \n, \s, etc.) and handle separation from commas as well. A vocabulary for each dataset is created by selecting the most frequent words. A lookup table maps each word to an index within the vocabulary range. The prompts and responses are then encoded as tf.long tensors, clipped to length 1000 using these indices so that they can be passed to an embedding layer before moving through the transformer architecture.

Image-to-Prompt Model

To generate the prompts, we use a transformer based decoder with keras built-ins. The model takes pre-encoded image features and captions as input, and the image features are passed through a dense layer to match up the size with the model's hidden size. The captions passed through a positional encoding layer and then are embedded using a word embedding layer. We included two layers of a simple encoder that applies multi-headed attention and a feed forward network to process the caption embeddings. The transformed caption embeddings and image features are then passed to a transformer decoder, which produces the output logits for generating captions. The final classification layer predicts the next word in the sequence.

During the process of developing this model, we experimented with adding learnable positional embeddings to capture sequential information, but this ultimately dropped validation accuracy and increased validation perplexity, so we elected not to include this. We also experimented with adding/removing the LSTM from the encoder layers. Removing the LSTM sped up the training runtime, but ultimately did not yield better captioning results.



To train, we used an H100 GPU over 20 epochs with early stopping. The learning rate was 1e-4 with a batch size of 64.

Prompt-to-Story Model

Originally, for the prompt-to-story generation architecture, we planned to use a two-level neural network approach originally explored in an existing research paper (Fan et. al. 2018). This method divides story generation into two distinct phases: the “planner” and the “generator.” The planner converts a prompt, derived in our case from an image caption, into a sequence of sentence-level summaries. These summaries are then passed to the generator, which produces a full paragraph of natural language text. We chose this architecture based on the convincing results reported in the paper, where human judges often preferred the generated stories over those written by humans.

We built a story generator from scratch using a transformer-based sequence-to-sequence (seq2seq) model. The architecture consisted of an embedding layer, positional encoding, a multi-layer encoder-decoder structure, and a linear output layer. We trained for 10 epochs on a H100 with a learning rate of 1e-4 and a batch size of 16, hoping to get results similar to that of the paper.

Fine-tuned Gemma Model

In addition to training a model from scratch, we pulled Google’s open source Gemma 3 model weights from Hugging Face and fine-tuned the one-billion parameter version. The Gemma 3 1B model is a decoder-only, transformer-based model trained on a corpus of two-trillion tokens with a vocabulary size of approximately 262k tokens and a context window of 32k words [[Gemma](#)].

The Hugging Face ecosystem provides the ability to fine-tune models using LoRA (Low Rank Adaptation Architecture) [[LoRA](#)]. With LoRA, instead of having to conduct backpropagation on the entire one-billion parameter model, it is possible to create new trainable parameters of a much smaller size and inject them into the model. In particular, given an $n \times m$ weight matrix in the full model, W , LoRA creates two trainable matrices: an $n \times r$ matrix denoted A and an $r \times m$ matrix denoted B and updates that output of the layer to be $y = (W + AB)(x)$.

In our fine-tuning, we targeted all linear layers contained in the Gemma 3 architecture, training with a $r = 8$ hyperparameter, as suggested by Hugging face’s documentation [[Fine-tuning](#)]. We trained for 5 epochs on an H100 with a learning rate of 1e-4 and a batch size of 8.

Results

We relied on human judgment to assess the quality of the outputs because the loss function for transformer models in captioning and story generation tasks is often hard to interpret and doesn't always align with human-perceived quality. The standard metrics that we used to train the model, like sparse categorical cross-entropy, can miss important aspects like coherence, creativity, and fluency, so manual evaluation gave us a clearer sense of how well the model was performing in practice.

Image-to-Prompt Model

The results from our transformer model were underwhelming, with poor predictive performance and inconsistent validation metrics. Despite tuning efforts, the model struggled to generalize, suggesting that either the architecture or the data preprocessing needs improvement.

To improve performance, we experimented with several modifications: adjusting the vocabulary size, adding and removing an LSTM layer, using learnable positional encodings, and testing different numbers of encoder and decoder layers. While some of these changes slightly affected the results, none led to a meaningful or consistent improvement, indicating that deeper architectural or data-related adjustments may be needed.

What went wrong seems to boil down to a few key issues: the data may have been too noisy or insufficient for the task, the loss function proved difficult to optimize effectively, and the choice of vocabulary size may have limited the model's ability to capture important patterns. Together, these factors likely contributed to the weak performance we observed.

Input Image	Generated Caption
	car yard uniforms puppy climb head head dance trail uniforms throws people day throws snowboarding throws snowboarding chasing gym past

	frisbee picture off face jersey jersey jersey gym gym gym gym gym gym chases animal soccer gym gym gym gym gym rocky
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To improve our image-to-caption transformer model, we could start by using better image encodings, such as CLIP or ViT, to provide richer visual representations. Incorporating visual attention mechanisms may also help the model focus on relevant parts of the image during caption generation. Additionally, training on a larger and higher-quality dataset could improve generalization and output quality. Finally, using decoding strategies like beam search might enhance the fluency and relevance of the generated captions.

Prompt-to-Story Model

The results from this model were extremely poor, likely due to a masking error in the original architecture. The output of the model is nonsensical, though it tends to predict tokens which appear frequently in the dataset.

Input Prompt: You walk a lonely road.	Model Response: you walk a lonely road <newline> , . <newline> of her a his , a <newline> `` , i it i to i i . . in i had , and but , of i the `` i and " the the <newline> . the on . . the i the not to . my he <newline> . , he . you my , it<STOP>
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Due to the performance of this model, we decided to transition to fine-tuning the Gemma 3 model's one-billion parameter version.

Fine-tuned Model

Fine-tuning can quickly alter the tone and behavior of foundation models. The fine-tuned Gemma model adheres more closely to the style of story in our web-scraped dataset. For example, it generates dialogue far more frequently than the base model, and it also provides responses which break the base model's trained behavior, such as refusing to engage in profanity.

Despite this positive outcome, adjusting the base model led to catastrophic forgetting, which occurs when training a model on a new data lead to it forgetting how to perform tasks which it previously knew. In particular, our newly trained model fails to provide output that is as cohesive at the paragraph level as the original Gemma model. The model also displays a tendency to repeat itself and provide reduced sentence level complexity which were not present in the base model.

<i>Gemma 3, base model</i>	<i>Gemma 3, fine-tuned</i>
PROMPT: Write Season 30 of Game of Thrones.	PROMPT: Write Season 30 of Game of Thrones.
Response: The rain is relentless, a grey curtain clinging to the landscape. It's been three years since Jon Snow was slain, and the Iron Throne remains a monument to shattered hope. The North is fractured, ruled by warring lords—the Wildlings, House Stark, and the Greyjoys—all vying for control of dwindling resources. Winterfell is a fortress, but it's increasingly isolated, haunted by the ghosts of its past.	Response: "I just can't believe this is happening." "You think what's happening is happening?" "I'm telling you, I'm telling you! I mean, it's happened before. That's what happened with the White Walkers in the North, that's what happened to the North, and it's happened before with the Northern lands. You think I'm stupid? I'm telling you, this is what happens when the White Walkers show up." "That's the point—we're supposed to be telling a story, not getting into some fantasy shit ."

	"You don't understand. It's like... like the Wall of the North. The Wall of the North went up in the last episode, and it's been like this ever since. And it's not even just the North anymore. It's been going on for a while. Every town, every village in the kingdom is under siege."
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Integration

Here are the results of connecting both models together.

<i>Input Image</i>	<i>Generated Story</i>
 Caption: car yard uniforms puppy climb head head dance trail uniforms throws people day throws snowboarding throws snowboarding chasing gym past	"Don't be such a party animal! I've got to get down to the car yard. The car yard uniform is nice, I'll see you tomorrow." I stared at her as she walked to the door. I was wearing a full uniform from the car yard, I had to get my uniform cleaned out and I wasn't going to be caught short. I had a little trip to the gym to do, and a little party to be at. I threw my arms out to grab her, I do n't know what I would do without her.



Caption: frisbee picture off face jersey jersey
jersey gym gym gym gym gym chases animal
soccer gym gym gym gym rocky

Rocky was a strong dog. He was a great dog.
He was a good dog. He was a great dog.

His fur was thick and fluffy, and his eyes
were bright and black. He loved to chase.

Rocky loved the frisbee. He loved the way it
flew over the field and bounced off the grass,
and the way it smelled.

Rocky loved to run and run and run. He was
the best runner in the whole wide world. He
was the best dog in the whole wide world.

Rocky loved to chase. He loved to chase.

Unfortunately, the quality of linked models is limited by the weaker of the two components—if one model underperforms, the overall output tends to suffer. For example, in the first input output pair, the ground truth caption for an image was *"woman climbing in a helmet."* When this caption was passed to the story generator, it produced the following story:

"And I'm telling you, the rocks are too smooth for me to grip onto, and the wind is too strong for me to hold onto."

"Then why are you here?"

"You're just curious, aren't you?"

"Why?"

"You're curious. That's what you were when you decided to go rock climbing."

"You're not even going to look at me?"

"Yes, I am."

"Why are you so excited about this?"

Compared to the story generated from the caption produced by our image-to-prompt model, this story is more relevant to the image.

In conclusion, the relevance of the generated story to the image largely depends on the quality of the image caption. Similarly, the overall quality of the story produced by the generator is influenced by the quality of the writing prompts used for fine-tuning.

Challenges

One of the primary challenges we faced was the size and complexity of our datasets. We managed two different datasets – the image/caption dataset and the story dataset – which created more work. Implementing preprocessing steps for the Flickr8k dataset was difficult. In addition, working with the story dataset which was 2GB in size made training models difficult. For example, we ran out of memory multiple times on an H100 due to poor optimization for LoRA on large datasets. The story dataset’s size also slowed the process of developing the from scratch prompt-to-story model since training on a 2GB dataset is (unsurprisingly) a time consuming task.

Lastly, after the from scratch prompt-to-story model presents such poor results, there was little time to conduct extensive experimentation on the fine-tuned model. Unfortunately, we did not have the opportunity to adjust LoRA hyperparameters to obtain the correct balance between fitting to the story dataset and retaining the overall coherence of the base model’s output.

Reflection

We ultimately created a finished product that generates somewhat coherent short stories from image inputs, but the project didn’t fully meet all of our goals. While we were able to build a working pipeline, the image captioning model underperformed, and we weren’t able to build the full system from the ground up as we had initially hoped. Relative to our goals, we reached something close to our base target—producing an end-to-end system—but fell short of our stretch goals, particularly around designing a novel hierarchical story generation mechanism as we discussed in our initial proposal. Still, we were able to fine-tune an existing large language model, Gemma, for the story generation part, which allowed us to demonstrate interesting outputs despite the limitations upstream.

Our approach evolved significantly over time. We started with the plan of building our own hierarchical story generation model, with a high-level planner and low-level generator, but eventually pivoted to a simpler sequence-to-sequence model and then to fine-tuning Gemma. Although the inner workings of the architecture shifted, the overall structure of the pipeline of image → caption → story remained intact throughout the project. If we were to do the project over, we would probably have been more realistic early on about the time and resources needed to train two models and would have focused on either the captioning or the story generation component more deeply.

With more time, we see several areas for improvement. First, improving the quality of the captions would have an outsized impact on the entire system, since the pipeline was highly sensitive to the specificity and informativeness of the captions. Second, collecting better story data (such as stories from a source other than Reddit, which is not the most literary destination) would improve the quality of the stories produced by the prompt to story model. Third, continuing to build our own model for story generation would be an interesting way to explore more control over narrative structure and better tailor the generation process to our specific goals.

One of our biggest takeaways is that training and linking two generative models is significantly harder than expected, especially under time constraints. We also learned that it's important to leave more time than you think you need, particularly when working with ambitious, data-heavy goals. Another key insight is that in tasks like image captioning, high accuracy is inherently challenging because captions are subjective and image understanding is complex, especially when using relatively simple architectures. Even when our captioning model performed reasonably well on paper (by identifying key aspects of the scene etc.), the captions often lacked the nuance or specificity needed to serve as good prompts for the story generator. This experience underscored that our approach, while functional, may not be the best strategy for generating stories from images, and it pushed us to think critically about how better integrated or end-to-end approaches could help in the future.